t3-w09 — Need for statistic — Need for statistics: purposes, and the collection and tabulation of data

Run `diag-v3` · conditions C0 and C1 · the pack shown is draw 3 of each.

C0 `The curriculum objective alone. No lesson plan, no note.`

Input — what the model was sent

Contract `ComposeWriteObjectiveSection`, declaring `artefact_spec, lesson_context, unit_context`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Need for statistic

`subtopic`

Need for statistics: purposes, and the collection and tabulation of data

`term`

Third Term

`week`

9

`lesson_objectives`

• explain the purpose of statistic
• analyse and interpret statistical data presented in tables

Objective 1 — explain the purpose of statistic

`objective_text`

explain the purpose of statistic

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, current_step, instructional_events, student_tasks, activities, assessment_points, source_evidence, source_figures, objective_learner_state`.

Objective 2 — analyse and interpret statistical data presented in tables

`objective_text`

analyse and interpret statistical data presented in tables

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, current_step, instructional_events, student_tasks, activities, assessment_points, source_evidence, source_figures, objective_learner_state`.

Output — the pack it produced (draw 3)

Objective 1

Why Statistics Matter: Understanding Their Purpose

EXPLANATION

Statistics is the science of collecting, organising, presenting and interpreting data. It helps us answer questions such as:

* **What is happening?** – By looking at data we can see patterns, trends and relationships.
* **Why is it happening?** – Statistical analysis can suggest reasons behind observed patterns.
* **What might happen next?** – Using past data we can make predictions about the future.
* **How should we decide?** – Reliable data lets us choose the best course of action, whether it's planning a school event, improving a product, or shaping public policy.

In everyday life, statistics guides decisions in health, sport, business, government and many other fields. Understanding its purpose means recognising that data is a tool for informed, evidence-based choices.

WORKED_EXAMPLE

Worked Example:

Your school wants to know which sport is most popular among students so it can allocate funds for equipment.

1. **Collect data** – A questionnaire asks each student to name their favourite sport.
2. **Tabulate the results** – The responses are counted and placed in a table:

Sport	Number of votes
Football	45
Basketball	30
Athletics	20
Others	5

3. **Interpret the table** – Football has the highest count, so it is the most popular sport.
4. **Use the information** – The school decides to purchase new footballs and improve the football field, because the data shows this will benefit the greatest number of students.

This example shows how statistics helps us **understand preferences**, **make fair decisions**, and **allocate resources effectively**.

1. A local council wants to know the most common mode of transport to work among residents. Explain two reasons why collecting this data is useful.

answer: It helps the council plan transport infrastructure and reduces traffic congestion.

2. A bakery records the number of loaves sold each day for a week. How can this data help the bakery make decisions? List two purposes.

answer: It can identify the busiest days to manage staff and order supplies, and forecast future demand.

3. Scientists conduct a survey on the number of hours students study each week. What is one way this statistical information could be used?

answer: To identify study habits and develop programs to improve academic performance.

Objective 2

Understanding and Interpreting Data Tables

EXPLANATION

Statistics helps us organise, summarise and make sense of information. A common way to present data is in a table, where each column has a heading that tells you what the numbers represent, and each row gives the values for a particular item.

How to read a table

1. Identify the headings – they tell you what each column means.
2. Look at the rows – each row relates to one case (e.g., a sport, a subject, a household).
3. Find totals, percentages or comparisons by using the numbers in the rows and columns.
4. Ask yourself what the numbers tell you: which category is largest? What proportion does it represent? Are there any trends?

When you can answer these questions, you are analysing and interpreting the data.

WORKED_EXAMPLE

Worked Example

Table: Favourite sport among 75 students

Sport	Number of students
Football	30
Basketball	20
Athletics	15
Swimming	10

Interpretation

- **Most popular sport:** Football (30 students).
- **Total number of students:** 30 + 20 + 15 + 10 = 75.
- **Percentage preferring Football:** (30 ÷ 75) × 100 = 40%.
- **Percentage preferring Swimming:** (10 ÷ 75) × 100 ≈ 13.3%.

So, Football is the favourite sport, chosen by 40% of the class, while Swimming is the least popular, chosen by about 13% of the class.

1. Table: Favourite school subject among 120 students | Subject | Number of students | |---|---| | Mathematics | 45 | | English | 30 | | Science | 25 | | History | 20 | (a) Which subject is most popular? (b) What percentage of the whole class likes Mathematics?

answer: (a) Mathematics is most popular. (b) (45 ÷ 120) × 100 = 37.5%

2. Table: Daily water consumption (litres) for five households | Household | Litres per day | |---|---| | A | 150 | | B | 120 | | C | 180 | | D | 130 | | E | 170 | What is the average daily water consumption for these households?

answer: Average = (150 + 120 + 180 + 130 + 170) ÷ 5 = 750 ÷ 5 = 150 litres.

3. Table: Test score categories for 200 students | Grade | Number of students | |---|---| | A | 50 | | B | 70 | | C | 60 | | D | 20 | How many students scored C or lower (i.e., grades C and D combined)?

answer: Students with C or D = 60 + 20 = 80.

C1 `The objective plus the lesson plan's design: position, method labels, materials, entry behaviour and assessment.`

Input — what the model was sent

Contract `ComposeWriteObjectiveSection`, declaring `artefact_spec, lesson_context, unit_context`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Need for statistic

`subtopic`

Need for statistics: purposes, and the collection and tabulation of data

`term`

Third Term

`week`

9

`lesson_objectives`

• explain the purpose of statistic
• analyse and interpret statistical data presented in tables

`entry_behaviour`

• Students are familiar with counting and recording information.

`instructional_materials`

• classroom register, bottle tops, pencils or counters.

Objective 1 — explain the purpose of statistic

`objective_text`

explain the purpose of statistic

`current_step`

1

`instructional_events`

• present_content

`assessment_points`

• (a) Define statistic

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, student_tasks, activities, source_evidence, source_figures, objective_learner_state`.

Objective 2 — analyse and interpret statistical data presented in tables

`objective_text`

analyse and interpret statistical data presented in tables

`current_step`

2

`assessment_points`

• (b) State three methods of collecting data.
• (c) Prepare a frequency table for the following data: 1, 1, 3, 4, 4, 3, 2, 2, 3, 3, 4, 1, 2.

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, instructional_events, student_tasks, activities, source_evidence, source_figures, objective_learner_state`.

Output — the pack it produced (draw 3)

Objective 1

Why We Use Statistics

> EXPLANATION
> Statistics is a way of turning raw information into useful knowledge. By collecting data, organising it into tables or charts, summarising it with measures such as totals or averages, and interpreting what it tells us, we can answer questions, compare groups, spot trends, and make informed decisions. For example, a school might use statistics to see which subjects students enjoy most, or a shop might use them to decide which products to stock.

> WORKED_EXAMPLE
> Imagine you have a jar of 30 bottle caps: 12 red, 9 blue, and 9 green. By counting each colour (collecting data) and writing the numbers in a table, you can quickly see that red caps are the most common. This simple summary helps you understand the distribution of colours without having to look at each cap individually.

1. Define the term “statistics”.

answer: Statistics is the science of collecting, organising, summarising, and interpreting data.

2. List two main purposes of using statistics.

answer: 1) To describe data (summarise it). 2) To make predictions or informed decisions based on the data.

3. A class records the number of books each student reads in a month. Explain how statistics can help the teacher understand the reading habits of the class.

answer: By collecting the numbers, the teacher can organise them into a table, calculate totals and averages, find the most common number of books read, and identify any trends. This summary shows overall reading habits and highlights students who may need extra support or encouragement.

Understanding and Working with Statistical Tables

EXPLANATION

Statistics helps us make sense of information by organising it into tables, graphs and summaries.

Why we use statistics
- To describe what is happening (e.g., how many students prefer football).
- To compare groups (e.g., boys vs. girls).
- To predict future trends (e.g., expected rainfall).

How we collect data
1. **Surveys or questionnaires** – ask people for their answers.
2. **Observations** – watch and record what happens.
3. **Experiments** – carry out a test under controlled conditions.

Reading a table
A table lists *categories* (or values) in one column and the *frequency* – how many times each category occurs – in another column. By looking at the frequencies you can see which categories are most or least common.

Interpreting a frequency table
- Identify the highest frequency – the most common value (the mode).
- Compare frequencies to see patterns or gaps.
- Use the totals to calculate percentages if needed.

WORKED_EXAMPLE

Worked Example: Building a Frequency Table

Suppose you have the following data showing the number of books read by each student in a week: 2, 3, 2, 1, 4, 2, 3, 1, 2, 3.

1. List the possible values (1, 2, 3, 4).
2. Count how many times each value appears:
   - 1 appears **2** times
   - 2 appears **4** times
   - 3 appears **3** times
   - 4 appears **1** time
3. Write the frequency table:

```
Value | Frequency
------+----------
  1   |    2
  2   |    4
  3   |    3
  4   |    1
```

From this table you can see that reading **2 books** is the most common (the mode) and that only one student read **4 books**.

1. (b) State three methods of collecting data.
answer: 1. Surveys or questionnaires 2. Observations 3. Experiments

2. (c) Prepare a frequency table for the following data: 1, 1, 3, 4, 4, 3, 2, 2, 3, 3, 4, 1, 2.
answer: Value | Frequency 1 | 3 2 | 3 3 | 4 4 | 3

Result — per rubric dimension

Scores are 1–5, each the mean of 2 judge pass(es) on the same packet. The first three columns are the packs shown above (draw 3); the last is the lesson's mean over its 3 draws per condition — the study's number.

Rubric	C0	C1	C1 – C0, shown packs	C1 – C0, mean of 3 draws
Objective and curriculum alignment	4.00	4.50	**+0.50**	**+0.83**
Taught-lesson alignment	3.00	3.50	**+0.50**	**+0.83**
Scope and conceptual correctness	4.00	3.00	**-1.00**	**+0.67**
Student-facing clarity and usefulness	4.00	4.00	+0.00	**+0.33**